12TH CHEMISTRY · CHAPTER 21 · SOLVED EXERCISE

Halogenoalkanes

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS:

1 Encircle The Correct Answer.

I) Which of the following compound is a primary halogenoalkane?

(a) 1-bromo-2,2-dimethyl propane (b) 2-bromo-3-methyl butane
(c) 2-bromopropane (d) 2-bromo-2-methyl propane

Answer: (a) 1-bromo-2,2-dimethyl propane — In $(CH_3)_3C–CH_2–Br$ the carbon bearing bromine is attached to only one other carbon atom.

II) The reactivity order of halogenoalkanes for a particular alkyl group is:

(a) Fluoride > Chloride > Bromide > Iodide
(b) Chloride > Bromide > Fluoride > Chloride
(c) Iodide > Bromide > Chloride > Fluoride
(d) Bromide > Iodide > Chloride > Fluoride

Answer: (c) Iodide > Bromide > Chloride > Fluoride — The C–I bond is the weakest (228 kJ/mol) and the C–F bond the strongest (467 kJ/mol).

III) S_N2 reactions can be best carried out with:

(a) Primary halogenoalkanes (b) Secondary halogenoalkanes
(c) Tertiary halogenoalkanes (d) Halogenoarenes

Answer: (a) Primary halogenoalkanes — Their electrophilic carbon is least crowded, so the nucleophile can attack it easily.

IV) Elimination bimolecular reactions involve:

(a) First order kinetics (b) Second order kinetics
(c) Third order kinetics (d) Zero order kinetics

Answer: (b) Second order kinetics — Rate = k [Alkyl halide][Base]; order = 1 + 1 = 2.

V) For which mechanisms, the first step involved is the same:

(a) E1 and E2 (b) E2 and S_N2
(c) S_N1 and E2 (d) E1 and S_N1

Answer: (d) E1 and S_N1 — In both, the first step is the slow ionization of the substrate to give a carbocation.

studysapiens.com

VI) Halogenoalkanes are considered to be very reactive compounds towards nucleophiles, because:

(a) They have an electrophilic carbon
(b) They have an electrophilic carbon and a good leaving group
(c) They have an electrophilic carbon and a bad leaving group
(d) They have a nucleophilic carbon and a good leaving group

Answer: (b) They have an electrophilic carbon and a good leaving group — The δ+ carbon attracts the nucleophile and the halide ion leaves easily.

VII) 2-chloropropane will follow which mechanism when it reacts with aqueous KOH resulting in the formation of an alcohol.

(a) S_N1
(b) S_N2
(c) Mixed S_N1 and S_N2
(d) E1

Answer: (c) Mixed S_N1 and S_N2 — It is a secondary halogenoalkane, which can follow both S_N2 and S_N1.

VIII) Which one of the following is not a nucleophile?

(a) H_2O (b) H_2S
(c) BF_3 (d) NH_3

Answer: (c) BF_3 — BF_3 is electron deficient and has no lone pair on boron to donate; it is an electrophile.

IX) When aqueous solution of alkyl halides are reacted with $AgNO_3$, the most rapid ppt. formation occurs with:

(a) fluoroalkane (b) chloroalkane
(c) bromoalkane (d) iodoalkane

Answer: (d) iodoalkane — The C–I bond is the weakest, so the iodide ion is released most quickly.

X) The rate of E1 reaction depends upon:

(a) The concentration of the substrate
(b) The concentration of the base
(c) The concentrations of substrate as well as base
(d) Neither the concentration of substrate nor the base

Answer: (a) The concentration of the substrate — Only the alkyl halide takes part in the slow, rate-determining step.

MCQ II option (b) is printed as "Chloride > Bromide > Fluoride > Chloride" (chloride twice); it is reproduced as printed.

Q2. SHORT ANSWER QUESTIONS

a) Why halogenoalkanes are known to be very reactive compounds?

The C–X bond is polar because the halogen is more electronegative than carbon. The carbon carries a partial positive charge and acts as an electrophilic centre that is easily attacked by nucleophiles, and the halide ion is a good leaving group. A base can also remove the hydrogen from the adjacent carbon. So halogenoalkanes readily undergo both nucleophilic substitution and elimination reactions.

b) Why fluoroalkanes rarely undergo nucleophilic substitution reactions?

The reactivity of a halogenoalkane depends mainly on the strength of its carbon–halogen bond. The C–F bond is the strongest (467 kJ/mol), so it is very difficult to break. It is so strong that fluoroalkanes do not react under ordinary conditions.

c) Why primary halogenoalkanes undergo substitution reactions by S_N2 mechanism?

For steric reasons. In a primary halogenoalkane the carbon bearing the halogen is attached to hydrogen atoms or only one carbon atom, so it is not crowded. The nucleophile can easily approach and attack this electrophilic carbon from the side opposite to the leaving group in a single step. Also, a primary carbocation is not stable, so ionization does not occur.

d) Does 1-bromo-2,2-dimethylpropane undergo elimination reaction or not? Justify.

No. Its structure is $(CH_3)_3C–CH_2–Br$. β-Elimination needs a hydrogen atom on the β-carbon (the carbon next to the one bearing the halogen). Here the β-carbon is attached to three methyl groups and has no hydrogen atom, so a base cannot remove a β-hydrogen and no alkene can form.

e) Why tertiary halogenoalkanes undergo elimination reaction by E1 mechanism?

In E1 the first, rate-determining step is the ionization of the substrate to give a carbocation. A tertiary halogenoalkane gives a tertiary carbocation, which is the most stable carbocation (three alkyl groups with +I effect), so it forms easily. The base then removes a β-hydrogen in the fast second step to give the alkene.

f) Hydroxide ion is a good base as well as a good nucleophile. Justify.

A halogenoalkane reacts with sodium hydroxide to give both substitution and elimination products, depending on the conditions. The hydroxide ion is both a good nucleophile and a good base. As a nucleophile it replaces the halogen atom to form an alcohol; as a base it removes the β-hydrogen and the halogen to give an alkene. The reaction is carried out at higher temperature under reflux, to avoid the loss of volatile products.

$$CH_3–CH_2–Br + NaOH \rightarrow CH_3–CH_2–OH + NaBr \text{ (aqueous; substitution)}$$

$$CH_3–CH_2–Br + NaOH \rightarrow CH_2=CH_2 + H_2O + NaBr \text{ (alcohol; elimination)}$$

g) What is the role of iron chloride ($FeCl_3$) during the chlorination of benzene?

Anhydrous $FeCl_3$ is the catalyst (a Lewis acid, halogen carrier). It polarises the chlorine molecule and generates the electrophile Cl^+, which attacks the benzene ring in an electrophilic aromatic substitution. It is regenerated at the end.

$$Cl_2 + FeCl_3 \rightarrow Cl^+ + [FeCl_4]^-$$

$$C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl\ (FeCl_3)$$

h) Which among haloalkanes or haloarenes, is more reactive towards nucleophiles?

Haloalkanes are more reactive. Their carbon attached to the halogen is electrophilic and the C–X bond is an ordinary single bond. In haloarenes the lone pair of the halogen is delocalised into the ring, giving the C–X bond partial double-bond character; the bond is stronger and difficult to break.

i) Write the equation for the reaction of 1-chloropropane with $AgNO_3$.

$$CH_3–CH_2–CH_2–Cl + H–OH \rightarrow CH_3–CH_2–CH_2–OH + H^+ + Cl^-$$

$$Ag^+ + Cl^- \rightarrow AgCl\ (\text{white ppt})$$

$$\text{Overall: } CH_3CH_2CH_2Cl + AgNO_3 + H_2O \rightarrow CH_3CH_2CH_2OH + AgCl + HNO_3$$

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Predict the mechanism which the chloroalkane will adopt while giving nucleophilic substitution reactions.

The mechanism depends on the type of chloroalkane:

Chloroalkane	Mechanism	Reason
Primary, e.g., chloroethane	S_N2	The carbon is not crowded, so the nucleophile attacks easily; a primary carbocation is unstable.
Secondary, e.g., 2-chloropropane	Both S_N2 and S_N1	Depends on the nature of the solvent.
Tertiary, e.g., 2-chloro-2-methylpropane	S_N1	A stable tertiary carbocation is formed; three bulky groups block back-side attack.

b) Why nucleophilic substitution and elimination reactions of haloalkanes proceed together under similar conditions?

A halogenoalkane has two sites that can be attacked: the electrophilic carbon of the C–X bond, and the hydrogen on the adjacent (β) carbon. Many reagents, such as OH^-, are both good nucleophiles and good bases. Acting as a nucleophile the reagent attacks the carbon and gives the substitution product; acting as a base it removes the β-hydrogen and gives the alkene. So the two reactions take place simultaneously and compete with each other. The conditions decide the major product: aqueous NaOH favours substitution, alcoholic NaOH favours elimination.

c) It is difficult for chlorobenzene to undergo nucleophilic substitution reactions but it is easier for it to undergo electrophilic substitution reactions. Give reasons.

Nucleophilic substitution is difficult: the lone pair of chlorine is delocalised into the aromatic ring, so the C–Cl bond has partial double-bond character. It is stronger and difficult to break. The electron-rich ring also repels the incoming nucleophile.

Electrophilic substitution is easier: the benzene ring has a cloud of delocalised π electrons, which attracts electrophiles. An electrophile replaces a ring hydrogen and the stable aromatic system is restored, as in the reactions of benzene.

d) Differentiate between E2 and E1 reactions. Draw a table for your response.

E2	E1
Single step.	Two steps.
Base removes the β-hydrogen and the leaving group leaves at the same time.	Slow ionization gives a carbocation; then the base removes a β-hydrogen (fast).
No intermediate.	Carbocation intermediate.
Bimolecular; second order: Rate = k[Alkyl halide][Base].	Unimolecular; first order: Rate = k[Alkyl halide].
Generally followed by primary halogenoalkanes.	Followed by tertiary halogenoalkanes.
Example: bromoethane → ethene.	Example: 2-bromo-2-methylpropane → 2-methylpropene.

e) Why does CH_3-CH_2-I give a precipitate with aqueous silver nitrate earlier than CH_3-CH_2-Cl?

The precipitate forms only after the C–X bond has been broken by water to release the halide ion. The C–I bond (228 kJ/mol) is much weaker than the C–Cl bond (346 kJ/mol), so iodoethane is hydrolysed faster and releases I^- quickly, giving the yellow precipitate of AgI early. Chloroethane reacts very slowly, so the white precipitate of AgCl appears much later.

DESCRIPTIVE QUESTIONS

Q4. Describe how substitution of alcohol with following reagents help to synthesize halogenoalkanes i) KBr with H_2SO_4 ii) $SOCl_2$/Pyridine iii) PCl_3 iv) PCl_5

In each case the –OH group of the alcohol is replaced by a halogen atom.

i) KBr with H_2SO_4: the acid first reacts with potassium bromide to give hydrogen bromide, which then reacts with the alcohol to give the bromoalkane. The method is commonly used for bromoalkanes and iodoalkanes (H_3PO_4 is preferred for iodoalkanes, as it does not oxidize iodide ions to iodine).

$$CH_3–CH_2–OH + KBr + H_2SO_4 \rightarrow CH_3–CH_2–Br + H_2O + KHSO_4$$

ii) $SOCl_2$/Pyridine: thionyl chloride, with pyridine as catalyst, gives the alkyl chloride. The by-products HCl and SO_2 are gases which go off, leaving the pure product.

$$CH_3–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–Cl + SO_2 + HCl \text{ (pyridine)}$$

iii) PCl_3: phosphorus trihalides replace the –OH group by a halo group; three moles of alcohol react with one mole of PCl_3.

$$3CH_3–CH_2–OH + PCl_3 \rightarrow 3CH_3–CH_2–Cl + H_3PO_3 \text{ (heat)}$$

iv) PCl_5: phosphorus pentachloride reacts with the alcohol to give the chloroalkane.

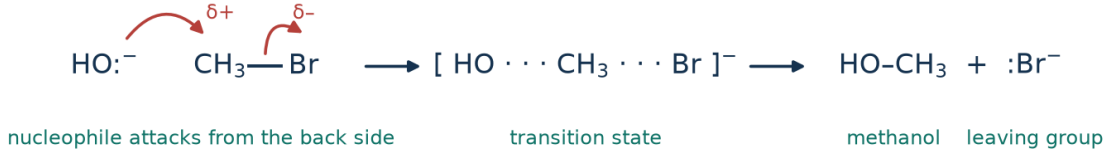
$$CH_3–CH_2–OH + PCl_5 \rightarrow CH_3–CH_2–Cl + POCl_3 + HCl$$

For phosphorus trihalides the textbook gives the equation with PBr_3; the PCl_3 equation is written on the same pattern.

Q5. Describe nucleophilic substitution reactions of halogenoalkane starting from bromoethane with following reagents: i) Aqueous NaOH ii) Ethanolic KCN iii) Ethanolic NH_3

i) Aqueous NaOH: the hydroxide ion acts as a nucleophile and replaces the bromine atom to form ethanol. The mixture is heated under reflux to avoid the loss of volatile products. Bromoethane is a primary halogenoalkane, so the mechanism is S_N2.

$$CH_3–CH_2–Br + NaOH \rightarrow CH_3–CH_2–OH + NaBr \text{ (aqueous, heat)}$$



The same single-step mechanism, shown for bromomethane

ii) Ethanolic KCN:

The cyanide ion (CN^-) is a strong nucleophile but a weak base, so it gives the substitution product, a nitrile. The halogenoalkane is heated under reflux with an ethanolic solution of potassium cyanide. Bromoethane gives propanenitrile (the chain becomes one carbon longer).

$$CH_3–CH_2–Br + KCN \rightarrow CH_3–CH_2–C{\equiv}N + KBr \text{ (alcohol, reflux)}$$

iii) Ethanolic NH_3:

Ammonia is a good nucleophile and reacts with halogenoalkanes favourably. Different types of amines can be produced if the amounts of halogenoalkane and ammonia are carefully controlled.



Ethanamine is also a nucleophile, so it competes with ammonia for the alkyl halide and the reaction continues:

$$NH_3 + C_2H_5Br \rightarrow C_2H_5NH_2 + HBr \text{ (ethanamine)}$$
$$C_2H_5NH_2 + C_2H_5Br \rightarrow (C_2H_5)_2NH + HBr \text{ (N-ethylethanamine)}$$
$$(C_2H_5)_2NH + C_2H_5Br \rightarrow (C_2H_5)_3N + HBr \text{ (N,N-diethylethanamine)}$$
$$(C_2H_5)_3N + C_2H_5Br \rightarrow [(C_2H_5)_4N]^+Br^- \text{ (quaternary ammonium salt)}$$

If a large excess of bromoethane is used, the final product is the quaternary ammonium salt. (The learning outcome states the conditions: NH_3 in ethanol, heated under pressure.)

Q6. Compare the reactivity of halogenoalkanes and halogenoarenes using chloroethane and chlorobenzene as examples.

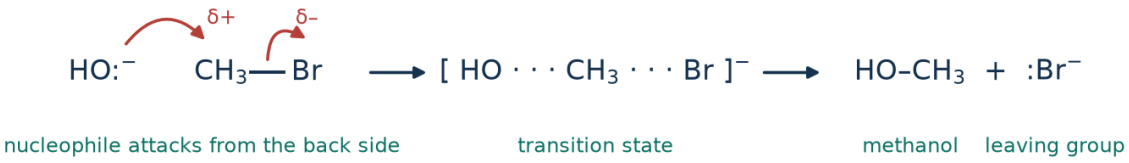
Halogenoalkanes (chloroethane)	Halogenoarenes (chlorobenzene)
Aliphatic: the halogen is attached to an alkyl group.	Aromatic: the halogen is attached directly to the benzene ring.
The carbon attached to the halogen has a partial positive charge (electrophile) and invites nucleophiles.	The lone pair of chlorine is delocalised into the aromatic ring.
C–Cl is an ordinary single bond, easily broken.	C–Cl has partial double-bond character; it is stronger and difficult to break.
Undergo nucleophilic substitution, and elimination with bases.	Less reactive towards nucleophiles; synthetically less useful.
$CH_3CH_2Cl + NaOH_{(aq)} \rightarrow CH_3CH_2OH + NaCl$	No reaction with NaOH under normal conditions.

Chlorobenzene instead undergoes electrophilic substitution in the ring, like benzene.

Q7. Describe the S_N1 and S_N2 mechanisms of nucleophilic substitution in halogenoalkanes.

Nucleophilic substitution on alkyl halides involves two main processes: the departure of the leaving group (breaking of the C–X bond) and the attack of the nucleophile (formation of the C–Nu bond). If the two processes occur simultaneously in a single step, the mechanism is **S_N2**. If the bond breaks in a first step, followed by formation of the new bond in a second step, the mechanism is **S_N1**.

Nucleophilic Substitution Bimolecular (S_N2) is a single-step mechanism. As soon as the nucleophile starts attacking the electrophilic carbon of the substrate, the bond to the leaving group starts breaking: the extent of bond formation is equal to the extent of bond breaking. The nucleophile attacks from the side **opposite** to the leaving group. Everything occurs at the same time, through a transition state.

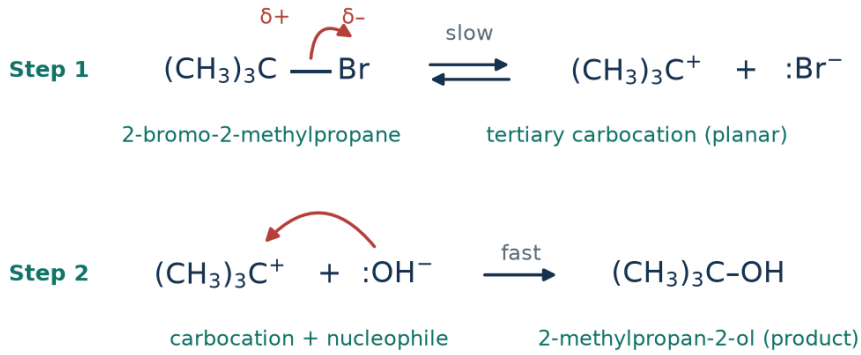


Since the reaction takes place in only one step, which is also the rate-determining step, and two molecules take part in it, it is a bimolecular reaction. The rate depends on the concentrations of both the alkyl halide and the nucleophile (second order):

$$\text{Rate} = k\,[\text{Alkyl halide}]^1\,[\text{Nucleophile}]^1 \text{ order} = 1 + 1 = 2$$

Primary monohalogenoalkanes always follow the S_N2 mechanism. This is due to steric reasons: the substrate carbon is attached to hydrogen atoms or only one carbon atom, so the nucleophile can approach and attack the electrophilic carbon easily.

Nucleophilic Substitution Unimolecular (S_N1) involves two steps. The first step is the reversible ionization of the alkyl halide in the presence of aqueous acetone or aqueous ethyl alcohol, giving a **carbocation** as an intermediate. In the second step, the carbocation is attacked by the nucleophile to give the product.



The first step involves breaking a covalent bond, so it is slow; the second step is the energetically favourable combination of ions, so it is fast. The first step is therefore the rate-determining step. Only one molecule takes part in it, so the mechanism is unimolecular and shows first order kinetics:

$$\text{Rate} = k\,[\text{Alkyl halide}]$$

Tertiary monohalogenoalkanes always follow the S_N1 mechanism. The major reason is the formation of the stable tertiary carbocation. Moreover, three bulky groups attached to the electrophilic carbon prevent the nucleophile from attacking from the back side of the leaving group. A **secondary** halogenoalkane can undergo both S_N2 and S_N1, depending on the nature of the solvent.

A carbocation is formed in the rate-determining step, so the more stable the carbocation, the more easily it is formed. Stability decreases as the positive inductive effect (+I) of the alkyl groups decreases:

$$(CH_3)_3C^+ > (CH_3)_2CH^+ > CH_3CH_2^+ > CH_3^+$$

tertiary > secondary > primary > methyl

S_N2	S_N1
Single step; bond making and bond breaking occur together.	Two steps; ionization first, then attack of the nucleophile.
Passes through a transition state; no intermediate.	A carbocation intermediate is formed.
Bimolecular; second order: Rate = k[RX][Nu].	Unimolecular; first order: Rate = k[RX].
Nucleophile attacks from the side opposite to the leaving group.	Nucleophile attacks the carbocation after the leaving group has gone.
Followed by primary halogenoalkanes (least steric hindrance).	Followed by tertiary halogenoalkanes (stable carbocation, crowded carbon).

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

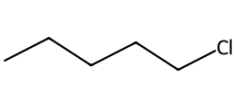
Quick Check 21.1

a) Write down the IUPAC name of the compound, $(CH_3)_2CH\text{-}CH_2\text{-}Cl$

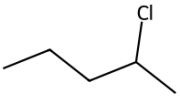
The longest chain has three carbon atoms, with chlorine on C-1 and a methyl group on C-2: **1-chloro-2-methylpropane** (a primary halogenoalkane).

b) Draw all possible structures of a compound having the molecular formula $C_5H_{11}Cl$. Classify each as primary, secondary or tertiary chloride.

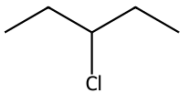
There are eight structural isomers: four primary, three secondary and one tertiary.



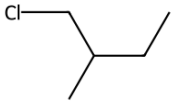
1-chloropentane (primary)



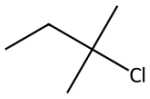
2-chloropentane (secondary)



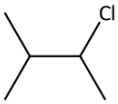
3-chloropentane (secondary)



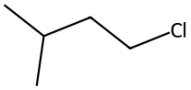
1-chloro-2-methylbutane (primary)



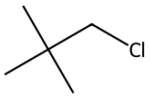
2-chloro-2-methylbutane (tertiary)



2-chloro-3-methylbutane (secondary)



1-chloro-3-methylbutane (primary)



1-chloro-2,2-dimethylpropane (primary)

Name	Condensed formula	Class
1-chloropentane	$CH_3CH_2CH_2CH_2CH_2Cl$	Primary
2-chloropentane	$CH_3CH_2CH_2CHClCH_3$	Secondary
3-chloropentane	$CH_3CH_2CHClCH_2CH_3$	Secondary
1-chloro-2-methylbutane	$CH_3CH_2CH(CH_3)CH_2Cl$	Primary
2-chloro-2-methylbutane	$CH_3CH_2CCl(CH_3)_2$	Tertiary
2-chloro-3-methylbutane	$(CH_3)_2CHCHClCH_3$	Secondary
1-chloro-3-methylbutane	$(CH_3)_2CHCH_2CH_2Cl$	Primary
1-chloro-2,2-dimethylpropane	$(CH_3)_3CCH_2Cl$	Primary

Quick Check 21.2

a) Give two methods to convert methanol into chloromethane.

1. With hydrogen chloride in the presence of $ZnCl_2$ as catalyst. **2.** With thionyl chloride in pyridine (PCl_5 can also be used).

$$CH_3\text{–}OH + HCl \rightarrow CH_3\text{–}Cl + H_2O\ (ZnCl_2)$$

$$CH_3–OH + SOCl_2 \rightarrow CH_3–Cl + SO_2 + HCl \text{ (pyridine)}$$

b) Give reactions of propanol with $SOCl_2$ and PBr_3.

$$CH_3–CH_2–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–CH_2–Cl + SO_2 + HCl \text{ (pyridine)}$$
$$3CH_3–CH_2–CH_2–OH + PBr_3 \rightarrow 3CH_3–CH_2–CH_2–Br + H_3PO_3$$

The products are 1-chloropropane and 1-bromopropane.

c) Can we prepare fluoromethane using methane and fluorine gas?

No, not in practice. Fluorine is the most reactive halogen and reacts with methane violently (explosively), even in the dark. The reaction cannot be controlled: it gives a mixture of polyfluorinated products, carbon and HF instead of pure fluoromethane.

The textbook gives halogenation of alkanes only for chlorine (and bromine); the behaviour of fluorine is general knowledge.

Quick Check 21.3

a) Why halogenoalkanes are considered as very reactive class of organic compounds?

The C–X bond is polar because the halogen is more electronegative than carbon. The carbon carries a partial positive charge and acts as an electrophilic centre that is easily attacked by nucleophiles, and the halide ion is a good leaving group. A base can also remove the hydrogen from the adjacent carbon. So halogenoalkanes readily undergo both nucleophilic substitution and elimination reactions.

b) Complete the following equation. $C_2H_5Br + CH_3COONa \rightarrow$

$$C_2H_5Br + CH_3COONa \rightarrow CH_3COOC_2H_5 + NaBr$$

The ethanoate ion (CH_3COO^-) is the nucleophile; it replaces bromine to give the ester ethyl ethanoate.

c) How will you convert ethane into ethanol and ethene?

First convert ethane into chloroethane; then use sodium hydroxide under two different conditions.

$$CH_3–CH_3 + Cl_2 \rightarrow CH_3–CH_2–Cl + HCl \text{ (light)}$$
$$CH_3–CH_2–Cl + NaOH \rightarrow CH_3–CH_2–OH + NaCl \text{ (aqueous: ethanol)}$$
$$CH_3–CH_2–Cl + NaOH \rightarrow CH_2=CH_2 + NaCl + H_2O \text{ (alcohol: ethene)}$$

d) Give the reaction of bromopropane with KCN. Is this a substitution reaction or an elimination reaction? Name the main product of this reaction.

$$CH_3–CH_2–CH_2–Br + KCN \rightarrow CH_3–CH_2–CH_2–C\equiv N + KBr \text{ (alcohol, reflux)}$$

It is a **nucleophilic substitution** reaction, because CN^- is a strong nucleophile but a weak base. The main product is **butanenitrile**.

Quick Check 21.4

a) What is the role of iron halide in the preparation of halogenoarene?

The iron halide (anhydrous $FeCl_3$ or $FeBr_3$) is a catalyst. It acts as a Lewis acid (halogen carrier): it polarises the halogen molecule and generates the electrophile X^+ ($X_2 + FeX_3 \rightarrow X^+ + [FeX_4]^-$), which attacks the benzene ring. It is regenerated at the end.

b) Chlorobenzene does not react with NaOH under normal conditions, but chloroethane does; explain why?

In chloroethane the carbon attached to chlorine carries a partial positive charge and the C–Cl bond is an ordinary single bond, so the hydroxide ion attacks this carbon and replaces chlorine. In chlorobenzene the lone pair of chlorine is delocalised into the aromatic ring, which gives the C–Cl bond partial double-bond character. The bond is stronger and difficult to break, so chlorobenzene does not react under normal conditions.

c) Can chlorobenzene undergo elimination? Explain why or why not?

No, not under normal conditions. The C–Cl bond is strong because of its partial double-bond character, so chloride does not leave easily. Also, removing H and Cl from two adjacent ring carbons would require a triple bond inside the six-membered ring, which would be highly strained and would disturb the stable aromatic system.

The textbook only states that the strong C–Cl bond makes halogenoarenes less reactive; the point about the ring is standard reasoning.

studysapiens.com